

ENVE 160B

Environmental Engineering Laboratory

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University of California, Riverside

<https://mdpetters.github.io/enve160b/>

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Syllabus

Course Title

ENVE-160B: Environmental Engineering Laboratory

Academic Department

Chemical and Environmental Engineering

Course Instructors

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Course Description

Consists of laboratory exercises in environmental engineering. Includes experiments in physical measurements, reaction kinetics, reactor analysis, and air pollution engineering. Emphasizes experimental design, analysis of results, and preparation of engineering reports.

Prerequisite(s): ENVE 133; or consent of instructor.

Meeting Time/Location

Lecture: M/W 12:30 PM - 3:20 PM

Location: Riverside Campus | Bourns Hall | Room B134

Discussion: F 3:00 PM - 3:50 PM

Location: Riverside Campus | Skye Hall | Room 172

Course Objectives

This course introduces students to concepts essential for making and analyzing measurements relevant to environmental engineering. It is intended to help train students to design and conduct reliable experimental measurements. The main objectives of the course are to apply the fundamental concepts developed in previous coursework, to set up and conduct complex experiments, and to support the theories formulated in lecture courses. The experiments give students an opportunity to think creatively, apply the skills developed in lecture courses, and communicate effectively through written and oral reports.

🔗 Learning Outcomes

1. Gain hands on experience with air quality measurement techniques
2. Engage in design and conduct of open-ended experiments
3. Gain familiarity with reporting methods of environmental data
4. Be able to put measurements into perspective based on results available in the literature
5. Work in complex team environment to accomplish significant tasks
6. Employ mathematical and computational techniques in interpretation of collected data
7. Gain experience presenting results in group setting and in participating in an active discussion aimed at encouraging the different groups to integrate and interpret their results.

ABET Student Outcomes

The Accreditation Board for Engineering and Technology (ABET) articulates critical student outcomes that support the program educational objectives. Attainment of these outcomes prepares graduates to enter the professional practice of engineering. The table below summarizes how this course meets various ABET targets.

	ABET Criteria	This Course
1.	An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	Open-ended prompts and requirement to engage with the peer-reviewed literature force students to identify, formulate and solve complex engineering problems in the context of engineering science and mathematics.
2.	An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	Students design and carry out experiments that target public health and environmental welfare.
3.	An ability to communicate effectively with a range of audiences.	Students learn to communicate findings via oral and written reports that meet current professional standards. Students learn to participate in active discussions with peers and an audience.
4.	An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	Students learn to effectively collaborate in a team, be accountable as both individuals and as a group for produced results.
5.	An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	Students learn how to conduct hand-on experiments through multiple labs. They analyze and interpret the data and are required to draw conclusions from the data in presentations and reports.
6.	An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.	Students are provided lab manuals, auxiliary data websites, and peer-reviewed manuscripts as a guide. They are required to acquire and apply this knowledge using their own learning strategies.

Course Schedule

M	Activity	W	Activity	F	Activity	Report Due
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		Sep 23		Sep 25	<i>no meeting</i>	
Sep 28	Syllabus/Safety	Sep 30	<i>no meeting</i>	Oct 2	<i>no meeting</i>	
Oct 5	Rotation A Session 1	Oct 7	Rotation A Session 1	Oct 9	Reporting	
Oct 12	Rotation B Session 1	Oct 14	Rotation B Session 1	Oct 16	Reporting	
Oct 19	Rotation A Session 2	Oct 21	Rotation A Session 2	Oct 23	Reporting	Monday: Rotation A Report #1
Oct 26	Rotation B Session 2	Oct 28	Rotation B Session 2	Oct 30	Reporting	Monday: Rotation B Report #1
Nov 2	Make-up	Nov 4	Make-up	Nov 6	Reporting	
Nov 9	Rotation A Session 3	Nov 11	Rotation A Session 3	Nov 13	Reporting	Monday: Rotation A Report #2
Nov 16	Rotation B Session 3	Nov 18	Rotation B Session 3	Nov 20	Reporting	Monday: Rotation B Report #2
Nov 23	Make-up	Nov 25	Make-up	Nov 27	<i>Thanksgiving break</i>	

| Nov 30 | Make-up | Dec 2 | Make-up | Dec 4 | *no meeting* | Monday: Rotation A/B Report #3 |

Assessment

Assessment of learning objectives will be through participation and laboratory reports.

Evaluation of Student Performance

The grade components in the class include:

Activity	Weight	Details
Reports	10%	Prelab Quizzes.
Reports	90%	Three (3) laboratory reports. Laboratory report rubric factors in attendance.

Laboratory Reports Each **individual** will turn in a report for each of the three labs that summarizes the experimental design and results. The grade assigned to each student will be based on a rubric that includes prelab reports, attendance, and quality of the written report. Details are provided in the [Lab Reports](https://mdpetters.github.io/enve160b/labreport/) <<https://mdpetters.github.io/enve160b/labreport/>> section.

Participation Attendance in lab sessions and discussion is mandatory. Lab sessions marked as as make-up/group work are available for meetings of the groups with the instructor to discuss expectations and hypotheses for upcoming labs and findings and analysis for completed labs.

Late Homework Late submissions will receive no credit. An extension with partial credit *may* be granted if the student can document extenuating circumstances related to an extensive period of excused absence between the date that the homework assigned and the due date. In addition, *religious observances* or

other *unanticipated life events* may interfere with class attendance and homework submission. These *may* qualify for an extension, provided sufficient documentation is presented.

Lab Safety You need to follow the lab safety guidelines. If you are excluded due to improper preparedness (lack of personal protective equipment, lack of proper training you will not be allowed into the laboratory and will not receive attendance credit).

Definition of Letter Grades

This Course uses Letter Grading: A+, A, A-, B+, B, B-, C+, C, C-, D+, D, D-, F, I, W. Grading policies follow [UCR Regulations](https://senate.ucr.edu/regulations/section/513/01) <<https://senate.ucr.edu/regulations/section/513/01>>., Letter grades are defined as follows:

A - Distinction, indicating exceptional achievement. Mastery of course content at the highest level of attainment that can reasonably be expected of students at a given stage of development.

B - High pass, indicating more than minimal competence but less than exceptional achievement. Strong performance demonstrating a high level of attainment for a student at a given stage of development.

C - Pass, indicating competence. Totally acceptable performance demonstrating an adequate level of attainment for a student at a given stage of development.

D - Marginal pass. This grade is used to recognize that a student's performance was marginal but clearly better than that of students who receive an F.

F - Fail. This grade is used to indicate that the student has failed the course and will not receive any credit for the course. The F grade indicates that the student's performance in the required exercises has revealed almost no understanding of the course content.

I - Incomplete. This grade is used as a temporary grade. At the discretion of the instructor, students may be given an GD grade for work not completed because of a serious interruption in their work not caused by the student's negligence.

W - Withdrawal. Used on student's grade reports and transcripts to indicate all courses for which they have received official approval to drop or from which they have received official approval to withdraw after the deadlines for dropping.

Mastery Based Grading

This course will use mastery-based grading. Mastery based grading assigns letter grades "A" to "F" based on a point cutoff evaluated from the assignment rubric.

Note

Mastery based grading will never reduce your grade. If you earned an 95/100, you demonstrated mastery. The assignment difficulty is typically at a level that full credit can be achieved.

Under some circumstances I *may* allow resubmission of an assignment for you to demonstrate mastery.

Absence Policy

Consistent class attendance is the number one predictor of academic success. Therefore, class attendance is required. Excused absences include

1. Medical Conditions
2. Death in the Immediate Family
3. Military Orders
4. Legal Obligations

Written verification from a from physician, lawyer, judge, or commanding officer needs to be provided for any makeup exam.

Methods of Student Instruction

- Laboratory Session: Hands on work to collect measurements and interact with the instructors. Used for developing measurement plans and working out hypotheses.
- Report Writing: Students submit reports and obtain timely feedback to help improve future reports.
- Discussion Sessions: Discussion sessions are used for group presentations.

Readings and Recommended Texts

Reading material is linked to each lab session.

Academic Accommodations

Students are encouraged to take an active role as their own advocates. They are not required to identify themselves as having a disability; however, students are not eligible for accommodations unless they self-identify. Students are responsible for visiting the SDRC office (Student Disability Resource Center, 125 Costo Hall (951) 827-4538, <http://sdrc.ucr.edu/>, and obtaining and delivering letters detailing appropriate accommodations to each of their instructors in a timely manner every quarter.

Instructors are responsible for assisting in providing accommodations; students are required to make requests in an appropriate and timely manner. Ordinarily, required accommodations are described in a letter, which is given to the instructor by the student. Classroom accommodations may involve providing space for a sign language interpreter or real-time captionist, asking students to clear a path for wheelchair users, announcing need for note takers, for example. Some accommodations, such as materials in Braille, large print, or electronic format, readers, and special equipment for testing or composing papers, are provided outside of the classroom by SDRC. UCR [practices for students](https://studentdocs.ucr.edu/dos/uc-riverside_dean-of-studetns_practices-for-students-with-disabilities.pdf) <https://studentdocs.ucr.edu/dos/uc-riverside_dean-of-studetns_practices-for-students-with-disabilities.pdf> summarized by the Dean.

Academic Rights and Responsibilities

All students, faculty, and staff are responsible for understanding and complying with the University's stated academic requirements. Students should feel free to express their thoughts and opinions in an academic forum. Assignments must be completed by the student for whom the work is assigned and without unauthorized aid of any kind.

Academic Integrity

At UCR we are committed to upholding and promoting the values of the Tartan Soul: Integrity, Accountability, Excellence, and Respect. As a student in this class, it is your responsibility to act in accordance with these values by completing all assignments in the manner described, and by informing

the instructor of suspected acts of academic misconduct by your peers. By doing so, you will not only affirm your own integrity, but also the integrity of the intellectual work of this University, and the degree which it represents. Should you choose to commit academic misconduct in this class, you will be held accountable according to the policies set forth by the University, and will incur appropriate consequences both in this class and from Student Conduct and Academic Integrity Programs. For more information regarding University policy and its enforcement, please visit: <https://conduct.ucr.edu/>.

⚠ Misconduct

The use of generative AI software such as ChatGPT or GitHub Copilot is expressly forbidden in this course. Use of these tools will be treated as academic misconduct.

Cheating is the giving, taking, or presenting of information or material by a student that unethically or fraudulently aids oneself or another person on any work which is to be considered in the determination of a grade or the completion of academic requirements or the enhancement of that student's record or academic career. Cheating includes, but is not limited to the following actions:

1. Copying from someone else's assignment, examination, or other academic exercise.
2. Possessing, buying, selling, removing, receiving, or using, at any time or in any manner not prescribed by the faculty member, any information related to an instrument of academic evaluation.
3. Using materials, equipment, or assistance in connection with an assignment, examination, or other academic exercise which have not been authorized by the faculty member, including but not limited to, notes, calculator, or other technology.
4. Obtaining or attempting to obtain, in a dishonest manner, any material relating to a student's academic work.
5. Working with another or others in completing an assignment, examination, or other academic exercise when the faculty member has required independent and unaided action.
6. Attempting to influence or change an academic evaluation, grade, or record by unfair means.
7. Permitting another individual to substitute for one's self in an academic evaluation.
8. Marking or submitting an examination or evaluation material in a manner designed to deceive the grading system.
9. Failing to comply with a specific condition of academic integrity which has been clearly announced in a particular course.
10. Submitting, without prior permission of the faculty member, any work by a student which has at any time been submitted in identical or similar form by that student in fulfillment of any other academic requirement at any institution.
11. Submitting of material in whole or part for academic evaluation that has been prepared by another individual(s).
12. Submitting data which have been altered or contrived in such a way as to be deliberately misleading.
13. Providing false information to the University in any manner to achieve an unfair advantage, enhance one's record, or complete a requirement.

Digital Course Components

Students may be required to disclose personally identifiable information to other students in the course, via digital tools, such as email or web-postings, where relevant to the course. Examples include online discussions of class topics, and posting of student coursework. All students are expected to respect the privacy of each other by not sharing or using such information outside the course.

Digital Course Components: Digital course components include

- Canvas Learning Management System

Non-Discrimination Policy

It is the policy of the University of California Riverside not to engage in discrimination against or harassment of any person employed or seeking employment with the University of California on the basis of race, color, national origin, religion, sex, gender, gender expression, gender identity, pregnancy, physical or mental disability, medical condition, genetic information, ancestry, marital status, age, sexual orientation, citizenship, or service in the uniformed services. Furthermore, University policy prohibits retaliation against any member of the University community filing a complaint of discrimination or harassment. This policy is intended to be consistent with the provisions of applicable State and Federal laws and University policies.

Mental Health

As a student, you may experience a range of issues that can cause barriers to learning, such as strained relationships, alcohol/drug problems, anxiety, depression, difficulty concentrating and lack of motivation. These mental health concerns or stressful events may lead to diminished academic performance or reduce a student's ability to participate in daily activities – UCR has resources to help! If you or a friend are in distress or experiencing any of these issues, please contact 951-UCR-TALK 24-hours a day to speak to a mental health professional, or call the National Suicide Prevention Lifeline at 1-800-273-TALK. Please refer to <https://mentalhealth.ucr.edu> for additional resources.

Basic Needs

If you or someone you know are in need of economic, food or housing support, you can find help at <https://basicneeds.ucr.edu>. You may be eligible for money to buy groceries via <https://basicneeds.ucr.edu/calfresh>. If you are in need of immediate assistance, see the Student Affairs Case Manager at <https://casemanagement.ucr.edu/> or visit the R'Pantry at <https://basicneeds.ucr.edu/rpantry>. You can find more information by clicking the Student Success Resources tab on your iLearn course menu.

Self Care

Categories of self care include biobehavioral (e.g., get plenty of sleep), affective-cognitive (e.g., watching a favorite, comforting movie), relational (e.g., talking with a friend), and spiritual (e.g., connecting with a faith community or the natural environment). In addition to the following resources, your instructor and TA are available to provide support and referrals.

University resources that may help you include:

- Counseling Center: (951) 827-5531 | [link <https://counseling.ucr.edu/>](https://counseling.ucr.edu/)
- Wellness Center: [link <https://well.ucr.edu/>](https://well.ucr.edu/)
- Campus Health Center: (951) 827-3031 | [link <https://campushealth.ucr.edu/>](https://campushealth.ucr.edu/)

- Student Special Services (for students with disabilities and veterans): [link](https://sdrc.ucr.edu/) <https://sdrc.ucr.edu/>
- Ethnic and gender program offices: [link](https://www.ucr.edu/students/ethnic_gender.html) <https://www.ucr.edu/students/ethnic_gender.html>
- Healing Highlanders Collegiate Recovery Program: [link](http://healinghighlanders.weebly.com) <http://healinghighlanders.weebly.com>
- Campus Safety Escort Service: (951) 827-3772
- UCR Ombudsperson: (951) 827-3213,
- UCR Sexual Assault Resource Services: (951) 827-6225
- UCR Student Conduct: (951) 827-4208
- UCR Student Affairs Case Manager: (951) 827-9354
- UCR Title IX/Sexual Harassment Office: (951) 827-5670
- Office of the Ombuds: (951) 827-3213
- Campus advocate for campus members affected by interpersonal violence:

Naddia Palacios | naddia.palacios@ucr.edu

384 and 386 Surge Building | (951) 827-6225 | [link](https://care.ucr.edu/) <https://care.ucr.edu/>

Off-campus resources include:

- Alternatives to Domestic Violence: (800) 339-7233
- Riverside Area Rape Crisis Center (24/7): (951) 686-7273)
- Riverside County Regional Medical Center: (951) 486-5650
- National Sexual Assault Hotline: (951) 656-4673
- National Dating Abuse Hotline: (866) 331-9474
- National Domestic Violence Hotline: (800) 799-7233
- Family Justice Center: (951) 955-6100

Safety Protocols

Prerequisite

Prior to the start of experiments, students will need to complete the safety training offered on through Canvas. Do not wait until last minute, the training process takes several days if there are access issues.

You will not be allowed to enter the laboratory if you have not uploaded certificates for

- Laboratory Safety Fundamentals
- Hazardous Materials & Waste Management

The lab is an inherently dangerous environment. While the experiments of the classes are designed to minimize the risk to students, there are hazards, both chemical and physical, present in the teaching labs. It is important to remember that the lab serves other student groups and classes and so there will be other chemicals present in the lab other than being used in 160B class experiments.

Note

Students must wear a protective laboratory coat and safety goggles during experimentation. Bring your own lab coat and goggles, there will be no available loaners. If you do not have a lab coat or goggles, you must obtain your own from the bookstore.

Because of their lack of protection, shorts and dresses (even when worn with tights) are not allowed in the laboratory. Legs must be covered to the foot. Pants that have holes in them will not be acceptable. To avoid exposure to hazardous materials, open-backed shirts, bare midriff shirts, or shirts which expose areas of the torso are not permitted. Shoes must cover the entire foot and be completely enclosed; perforated shoes, open-toed shoes, open-backed shoes, or high-heeled shoes are not permitted.

Caution

Students will not be allowed to enter the laboratory if not properly clothed. If you come not properly closed you will be asked to leave and not receive attendance credits for the lab

All persons in the laboratory shall wear eye protection with impact and splash protection whenever any chemicals or glassware are in use or on the benches anywhere in the laboratory. Splash goggles must be used when working with organic solvents (acetone) or corrosives greater than 1M in concentration. Safety glasses must be worn at all other times. PPE is required at all times in the lab, this includes coming to the lab to speak with the TAs or even at times when experiments are not being conducted. Always remove gloves and thoroughly wash your hands before exiting the laboratory. Students who are asked more than twice (in one lab period) to put their PPE on or follow any other safety procedure will be dismissed from the lab for that period.

Eating, drinking, chewing gum, applying cosmetics are prohibited in the laboratory. Beverage containers, cups, bottled water, and food containers are not permitted in the laboratory. These items must be kept outside or stowed in backpacks. Any food or beverage containers in the lab will be confiscated.

Good housekeeping is essential. Aisles, emergency exits, safety equipment must be unobstructed. Electric scooters, skateboards are not permitted to be stowed in the lab. Coats, bags, and other personal items should be stored in the proper areas in the laboratory room; not on the benchtops or in the aisles.

Clean up chemical spills (including water) immediately. Do not leave spilled chemicals on the benchtop or floor. Chemicals, including water, must be in labeled containers at all times. Keep glassware clean. Sinks should not be filled with dirty glassware; students are required to clean all items used and leave their workspace clean. TAs will check areas for cleanliness before students are dismissed from lab.

⚠ Caution

Do not dispose of waste chemicals in the sink drains or in the wastebaskets.

Containers for waste will be provided and must be used according to instructions from the TAs.

Laboratory Reports

Individual Assignment

Laboratory reports are **individual** assignments. After you have collected the data with your group, you must work on your own, producing your own analysis, graphs, and text. Copying code, text, or analysis from other group members is expressly forbidden.

Plagiarism

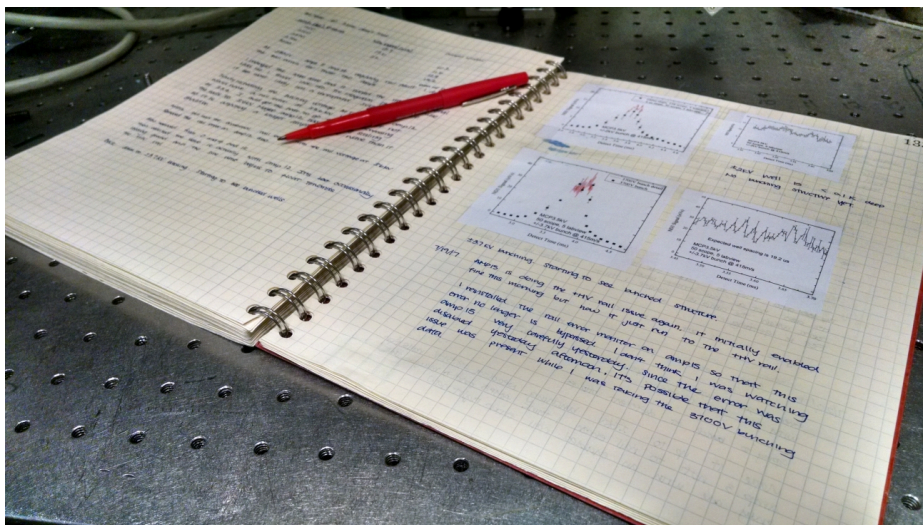
Please review the academic integrity section in the syllabus. Two important forms of academic misconduct relevant for the lab reports include

⚠ Misconduct

- The use of generative AI software such as ChatGPT or GitHub Copilot is expressly forbidden in this course.
- Verbatim copying or paraphrasing somebody else's report, paper, web-based entry, book, or any other source without use of quotation marks and citation (verbatim text) or citation (paraphrasing).

Reporting Mechanics

This course will require you to turn in hand-written reports using either pen or pencil. Reports are due on Monday at the beginning of the lab, at the date outlined in they syllabus. Reports will be graded and returned the same week on Wednesday so you can prepare the next report.



The rubric may require either hand-drawn figures or computer-generated figures. Computer-generated figures must be printed and pasted into the notebook. Lab reports will be submitted in the form of an exam blue book. Free blue books can be picked up at the beginning of the course.

Report Grade

Each report has a specific rubric that outlines the required sections and components of the grade. In general the components are

1. Prelab Quiz

You will be given a shorter in-person version of the online prelab quiz. The points will count to your report grade.

2. Attendance

Attendance points. You will receive attendance points for Monday and Wednesday lab sessions.

3. Reading Log

Each lab has a required reading associated with it. Before writing the report, provide a reading log. The reading log should be completed prior to starting the lab and contain the following.

- Figure/Data Interpretation: Select three one figures or tables/reaction from the text. What do you think this figure is showing? (Describe the variables, components, or process you observe.). What patterns, relationships, or trends do you notice?
- Mechanism: Choose one process, system, or reaction mechanism from the reading and explain how it works in your own words. Focus on cause-and-effect relationships or steps in the process.
- Mini Research: What result, claim, or observation in the reading surprised you? Find outside sources (scientific article, textbook, or reputable science site) related to this finding. What additional information did you learn?
- Reflection: Identify at least one concept that was challenging. What part of the reading was unclear? What specific question do you have about it?

3. Report

You will provide a hand-written report that is submitted on Monday at the beginning of the lab session on the date indicated in the syllabus. Detailed guidance is provided for each report. Don't forget to paste a copy of the grading rubric at the end of each report.

Rationale for Reporting Scheme

It may seem anachronistic that I request hand-written reports and that I take a strong stance against using AI. Writing by hand is difficult. I wrote the lab reports by hand and it made my hand hurt. It was significantly slower than writing it in a word processor. Computers and AI are powerful technologies that may appear to

make your life easier. However, there is growing evidence that computers and AI may do far more harm than doing good in education. While the following sections focus more on the harms of AI, some of the concerns also apply to just using word processors.

The main concern with AI in education (and beyond) is that when relying on AI, students will never learn critical skills like original thinking, writing, or coding. Even more concerning, there is emerging evidence that using generative AI causes underperformance at neural, linguistic, and behavioral tasks.¹ (Read, reduces your IQ). It is an existential threat to higher education because it facilitates plagiarism deployed at scale. Generative AI is highly addictive because it makes hard things seemingly easy. It provides an escape hatch for deadline pressure and the general feeling being overwhelmed by the difficulty of an assigned task. Yet, a student that is cutting and pasting images and text from manuals or ChatGPT into a word document is enrolled in a class in cutting and pasting class and not a university course. The bitter truth is that learning requires forcing ourselves to do hard things.

The cognitive science research and evidence from observing student behavior seems to be clear. Reducing the cognitive load makes your brain atrophy.¹ Conversely, taking handwritten notes, rather than typing, helps committing material to memory leads to higher achievement.² Students are human beings. Nothing will stop them from using ChatGPT and other shortcuts if they are available.³ For this reason it is time to force ourselves to slow down.

Writing by hand is thinking on paper.

Writing by hand has some amazing benefits. It requires us to first complete the analysis and organize the material before starting to write. Drafting figures with a pencil, and/or annotating graphs with pencil makes us pause and think about the data. Because handwriting is slower than typing, and less amenable to later reshuffling like we do in a word processor, it requires us to think more clearly about what we write. The assignments are **individual** because I want every student to have the benefit of fully understanding the experiment, analysis and implications. Because the reports are a significant amount of work, I have reduced the number of labs to three, thus freeing time for you to focus on analysis and reporting.

References

¹Kosmyna et al. (2025). Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task. <https://doi.org/10.48550/arXiv.2506.08872>.

²Flanigan, A.E., Wheeler, J., Colliot, T. et al. Typed Versus Handwritten Lecture Notes and College Student Achievement: A Meta-Analysis. *Educ Psychol Rev* 36, 78 (2024). <https://doi.org/10.1007/s10648-024-09914-w>.

³Clay Shirky: The Only Real Solution to the A.I. Cheating Crisis. *NY Times*
<<https://www.nytimes.com/2025/08/26/opinion/culture/ai-chatgpt-college-cheating-medieval.html>>.

Air Quality Refresher

Reviews the origin, temporal, and spatial distribution of criteria pollutants in the United States and Southern California.

Slides and Lecture Notes

Slide Deck: [link](https://docs.google.com/presentation/d/1PJGICqL3XkVluh1ZtjDrtL6Wts-7w8hutQsHf0oze0o/edit?usp=sharing) <https://docs.google.com/presentation/d/1PJGICqL3XkVluh1ZtjDrtL6Wts-7w8hutQsHf0oze0o/edit?usp=sharing>

Resources

Mates V Final Report ([link](https://www.aqmd.gov/docs/default-source/planning/mates-v/mates-v-final-report-9-24-21.pdf)) <https://www.aqmd.gov/docs/default-source/planning/mates-v/mates-v-final-report-9-24-21.pdf>

MATES Data Display ([link](https://experience.arcgis.com/experience/79d3b6304912414bb21ebdde80100b23/)) <https://experience.arcgis.com/experience/79d3b6304912414bb21ebdde80100b23/>

EPA Green Book ([link](https://www.epa.gov/green-book/green-book-map-download)) <https://www.epa.gov/green-book/green-book-map-download>

Photochemical Cycle of NO_2 , NO , and O_3

Learning Objectives

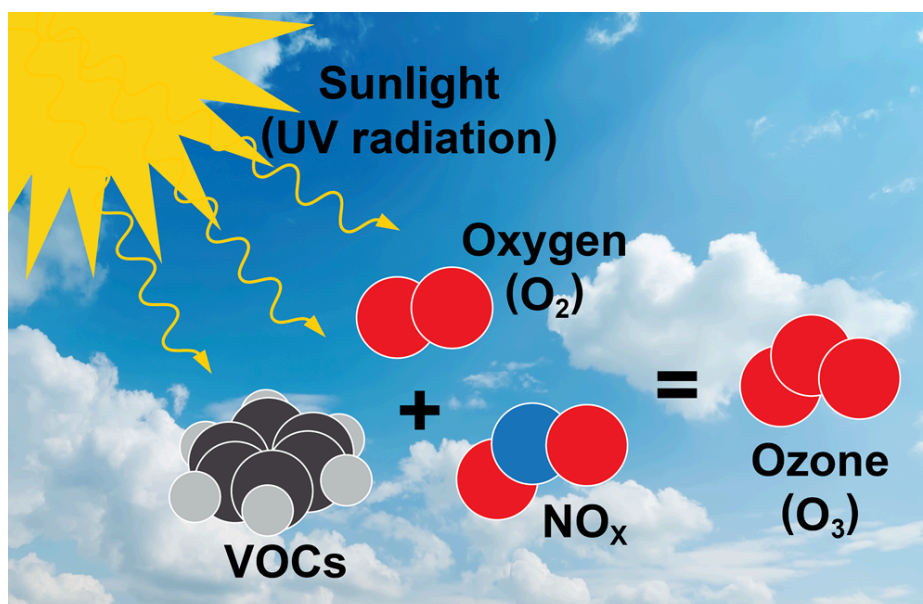
- Perform photochemical reactions inside environmental chambers.
- Explain the principle of operation of O_3 and NO_x gas-phase monitors.
- Explain the photochemical cycle of O_3 , NO , and NO_2 .
- Measure and calculate the ozone produced at the photostationary state.

Required Reading

Chapter 12, Ozone Air Pollution, from [Introduction to Atmospheric Chemistry](https://acmg.seas.harvard.edu/education-introduction-atmospheric-chemistry/)

<<https://acmg.seas.harvard.edu/education-introduction-atmospheric-chemistry/>>.

Background



Ozone is a gas composed of three atoms of oxygen (O_3). Ozone in the air we breathe can harm our health, especially on hot sunny days when ozone can reach unhealthy levels. People at greatest risk of harm from breathing air containing ozone include people with asthma. Elevated exposures to ozone can affect sensitive vegetation and ecosystems, including forests, parks, wildlife refuges and wilderness areas. In particular, ozone can harm sensitive vegetation during the growing season. (Source: [EPA](https://www.epa.gov/ground-level-ozone-pollution/ground-level-ozone-basics) <<https://www.epa.gov/ground-level-ozone-pollution/ground-level-ozone-basics>>). Tropospheric O_3 is generated from two major classes of precursor molecules: volatile organic compounds (VOCs) and nitrogen oxides NO_x , which is the sum of NO and NO_2 . Most of the direct emissions of NO_x are in the form of NO . Ozone levels exceeding

200 ppb are considered severe air pollution episodes. The current U.S. National Ambient Air Quality Standard for O_3 is an 8-hour average of 70 ppb. Understanding VOC chemistry is critical to explain the total ambient O_3 concentrations. Nevertheless, the photochemical cycle of NO_2 , NO , and O_3 remains the starting point for modeling tropospheric O_3 production.

Photochemical O_3 Production from NO_x

Sunlight at wavelength < 424 nm photolyzes NO_2 into NO and atomic O



The atomic O reacts with oxygen O_2 to form O_3 . Reaction (2) is the only source of atmospheric O_3 .



where M is a third body required to stabilize the excited product OO_2^* by collision. Finally O_3 reacts with NO to regenerate NO_2



Reactions (1)-(3) form the basic photochemical NO_x cycle. Cycling between NO and NO_2 takes place in the troposphere on a time scale of a minute in the daytime. There is no net production of O_3 , but some O_3 is present. The steady state O_3 concentration is

$$[O_3] = -\frac{1}{2} \left([NO]_0 - [O_3]_0 + \frac{j_{NO_2}}{k_3} \right) + \frac{1}{2} \left[\left([NO]_0 - [O_3]_0 + \frac{j_{NO_2}}{k_3} \right)^2 + 4 \frac{j_{NO_2}}{k_3} ([NO_2]_0 + [O_3]_0) \right]^{1/2} \quad (4)$$

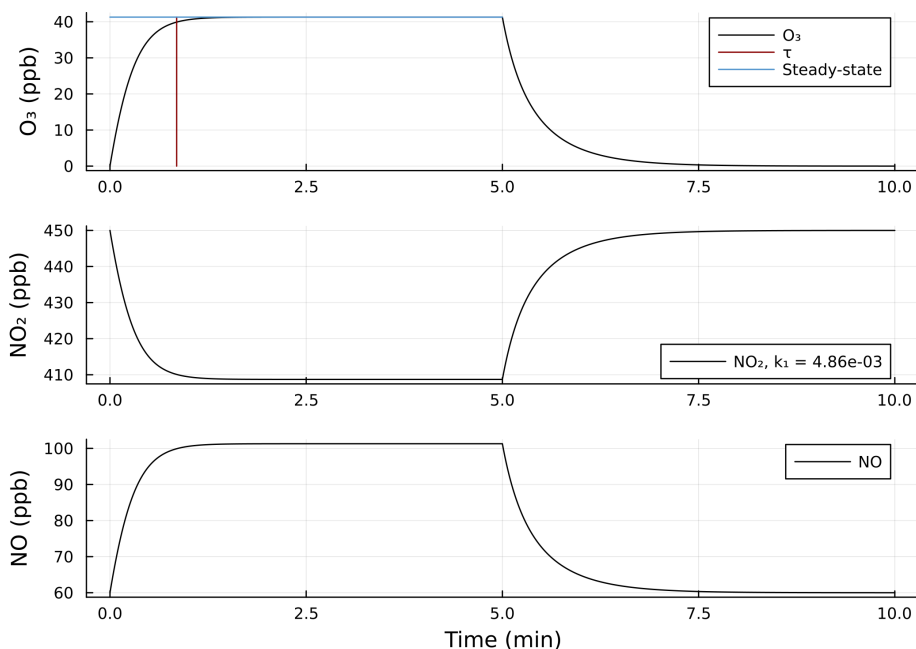
The ratio j_{NO_2}/k_3 depends on sunlight or blacklight intensity. For full blacklight intensity in the chamber used in this lab, $j_{NO_2}/k_3 \approx 10.2$ ppb. The characteristic relaxation time to steady state is

$$\tau = \frac{1}{k_3[NO]} \quad (5)$$

where $k_3 = 1.9 \times 10^{-14} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ at $T = 298K$.

Reaction Kinetics

The photolysis rate j_{NO_2} depends on the actinic flux (intensity of sunlight or intensity of blacklights). At noon in the cloud-free atmosphere $j_{NO_2} \approx 8 \times 10^{-3} \text{ s}^{-1}$ and otherwise lower. The value for $k_3 = 1.9 \times 10^{-14} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ at $T = 298K$. The figure below shows a typical evolution of NO_2 , NO , and O_3 for the experiment you are going to perform. At $t = 0$ the conditions are $[O_3]_0 = 0$ ppb, $[NO]_0 = 60$ ppb, and $[NO_2]_0 = 450$ ppb. The blacklights are turned on at $t = 0$. Predictions for the photostationary state concentration and relaxation time based on Eqs. (4) and (5) are provided. After 5 min, the lights are turned off and the system restores to the initial state.



Prompt

The goal is to measure the photostationary steady state of the NO_2 , NO , and O_3 system. The environmental chamber will be filled with zero air that is free of VOCs and other chemicals. A mix of $[NO]_0$, $[NO_2]_0$ and $[O_3]_0$ will be added to the chamber and the system will be allowed to equilibrate. Concentrations of NO_2 , NO , and O_3 will be monitored in real time using gas-analyzers. After the system is equilibrated, blacklights will be turned on. The blacklights provide photons $\lambda < 424 \text{ nm}$, thus initiating the photolysis of NO_2 and the production of O_3 . Observe the formation of O_3 until the photostationary state of the system is reached. Then turn off the light and observe the reaction of NO with O_3 , returning the system back to the initial state. Repeat the cycle at least once, preferably twice to ensure consistent results.

During the Lab

- Make sure the O_3 and NO_x gas analyzers are running. Make sure that the data acquisition system is turned on and recording data. Compare the computer logged data on the screen to the values reported on the analyzers. Fill the chamber with zero air. Verify that initial concentrations are approximately zero. Add the NO/NO_2 mixture to the chamber. Record the initial NO , NO_2 and O_3 concentrations. Turn on the light and let the system equilibrate to the photostationary steady state. Turn off the light and observe reversion to the initial state.
- Make a draft of the schematic drawing of the experimental setup. Carefully indicate (1) All of the instruments involved, including vacuum pumps, instruments, flow rates, flow paths, light sources (location and number), compressed air sources, and data acquisition system. (2) How the reactants are added to the chamber, (3) timings of reactants added and lights turning on and off. You should make a sketch during the experiment.
- Make a draft of the schematic drawing of the O_3 analyzer and discuss with your group and/or TA.
- Download the data from the computer for later analysis.

Report

Please follow the [report outline](https://docs.google.com/document/d/1IzfiSnkv9lwBkknXHKjJsLFzyCnfKwh2Q_X9xTLBZG4/edit?usp=sharing) <https://docs.google.com/document/d/1IzfiSnkv9lwBkknXHKjJsLFzyCnfKwh2Q_X9xTLBZG4/edit?usp=sharing>.

Don't forget to print and paste the [grading rubric](https://docs.google.com/document/d/13-KuUIEgavq18SuWSS4LJvFvVdt0BesU7G78qKYnscY/edit?usp=sharing) <<https://docs.google.com/document/d/13-KuUIEgavq18SuWSS4LJvFvVdt0BesU7G78qKYnscY/edit?usp=sharing>> at the end of your report. Not providing the grading rubric will lead to point deductions. Use the grading rubric as a checklist for how to prepare a proper report.

Methods

This lab uses an environmental chamber, a photometric UV absorption O_3 analyzer and a chemiluminescence NO_x analyzer.



The environmental chamber consists of a Teflon bag housed inside an enclosure that is lined with blacklights. The blacklights can be turned on/off using switches. An injection port is available to add reagents. You will receive a glass bulb that contains a nominal amount of NO . You will add the NO with the help of the TA and/or the instructor. Ozone is measured using a gas-analyzer. A 254 nm UV light signal is passed through the sample cell where it is absorbed in proportion to the amount of ozone present. Periodically, a switching valve alternates measurement between the sample stream and a sample that has been scrubbed of ozone. NO_x is measured using a second gas analyzer. The NO_x instrument determines the concentration of nitric oxide (NO), total nitrogen oxides (NO_x), the sum of NO and NO_2 and nitrogen dioxide (NO_2) in a sample stream. The principle of operation is [chemiluminescence](https://en.wikipedia.org/wiki/Chemiluminescence) <<https://en.wikipedia.org/wiki/Chemiluminescence>>. Chemiluminescence is the emission of light from a chemical

reaction and is triggered by the reaction of NO with ozone O_3 . The amount of light produced is linear with NO concentration. NO_2 is measured by converting NO_2 with to NO using heated molybdenum converter chip.

Resources

Teledyne Product Manuals (O_3 analyzer and NO_x analyzer) [link](https://www.teledyne-api.com/en-us/service-support/product-manuals) <<https://www.teledyne-api.com/en-us/service-support/product-manuals>>

Indoor Carbonyls

🔗 Learning Objectives

- Collect air samples for analysis by offline techniques.
- Explain the principle of operation of high performance liquid chromatography.
- Interpret chromatograms in the context of instrument standards to separate, identify and quantify components in a mixture.
- Contextualize the results using peer-reviewed studies

Required Reading

Chapter 4, pg. 1-15, from [Indoor Environmental Quality](https://doi.org/10.1201/9781420056747) <<https://doi.org/10.1201/9781420056747>>

Chapter 12.5 from [Analytical Chemistry](https://chem.libretexts.org/Bookshelves/Analytical_Chemistry/Analytical_Chemistry_2.1_by_David_Harvey/Analytical_Chemistry_2.1_(Harvey)/12%3A_Chromatographic_and_Electrophoretic_Methods/12.05%3A_High-Performance_Liquid_Chromatography)

<[https://chem.libretexts.org/Bookshelves/Analytical_Chemistry/Analytical_Chemistry_2.1_by_David_Harvey/Analytical_Chemistry_2.1_\(Harvey\)/12%3A_Chromatographic_and_Electrophoretic_Methods/12.05%3A_High-Performance_Liquid_Chromatography](https://chem.libretexts.org/Bookshelves/Analytical_Chemistry/Analytical_Chemistry_2.1_by_David_Harvey/Analytical_Chemistry_2.1_(Harvey)/12%3A_Chromatographic_and_Electrophoretic_Methods/12.05%3A_High-Performance_Liquid_Chromatography)>

DNPH Manual ([link](https://drive.google.com/file/d/1kUxEaHtLolt1s3gba3s77RTZVa4IFlb/view?usp=sharing)) <<https://drive.google.com/file/d/1kUxEaHtLolt1s3gba3s77RTZVa4IFlb/view?usp=sharing>>

Background

► **Video:** [HPLC](https://www.youtube.com/watch?v=nqRPRwO79kg) <<https://www.youtube.com/watch?v=nqRPRwO79kg>>

There is significant debate in the health and exposure community about the relative importance of indoor and outdoor air quality. While we most often think about emissions sources that directly impact outdoor air such as vehicle exhaust, industrial facilities, and wildfires, most of us spend much more time indoors than outdoors. Indoor air quality is influenced both by infiltration of outdoor air and by numerous indoor activities such as cooking and vacuuming, and numerous indoor emissions sources such as carpet, particle board, paint, cleaning products, and air fresheners. The relative importance of outdoor and indoor sources is impacted by individual preference and behavior (e.g., leaving windows open) and by increasingly common and effective energy savings measures incorporated in houses and buildings that reduce indoor/outdoor exchange of air. Carbonyls are one particular class of compounds that has garnered significant interest for indoor air quality. Several carbonyl compounds are known to pose significant health concerns when present at sufficiently high concentrations.

📺 Prompt

The objective is to separate, identify, and quantify carbonyls in indoor and outdoor environments. You will sample carbonyls at an indoor location and an outdoor location in close proximity, e.g. inside an

apartment and just outside. The samplers should be started and stopped at the same time. Sample for as close to 24 hours as possible. Record the actual start time and stop time and use the sampling duration in your analysis and not just the nominal 24 hours. You will decide and describe where the samples are collected. Indoor concentrations will likely be higher at a location that has new carpet, paint, furniture, etc. and where windows are typically closed.

Report

Please follow the [report outline](https://docs.google.com/document/d/1RQ76HCX6WOq7xpyWO7xmFcswf9AuZj-j-2FVD9GNivQ/edit?usp=sharing) <<https://docs.google.com/document/d/1RQ76HCX6WOq7xpyWO7xmFcswf9AuZj-j-2FVD9GNivQ/edit?usp=sharing>>.

Don't forget to print and paste the [grading rubric](https://docs.google.com/document/d/1-ccBXMLntglynwzMVW1I41nILHpoZQsjTngxwNQIVz8/edit?usp=sharing) <<https://docs.google.com/document/d/1-ccBXMLntglynwzMVW1I41nILHpoZQsjTngxwNQIVz8/edit?usp=sharing>> at the end of your report. Not providing the grading rubric will lead to point deductions. Use the grading rubric as a checklist for how to prepare a proper report.

Methods



You will be provided two DNPH cartridges in sealed pouches. DNPH cartridges consist of acidified 2,4-dinitrophenylhydrazine-coated silica particles that bind carbonyl containing gases. Air is forced to flow through the cartridge with the help of a pump. Carbonyls present in the air are trapped and extracted at a later time for chemical analysis. Be sure to label the sample bags and keep track of the two cartridges; mixing up the indoor and outdoor samples would obviously negatively impact your results. Also be sure to remove the cap on the cartridge. The samplers are started simply by connecting the pump electrical leads to one of the batteries you are provided. After stopping the samplers at the end of 24 hours, remove the DNPH cartridges, return them to their pouches, and keep them in a refrigerator until returning to campus on Wednesday.



► Video: [HPLC](https://www.youtube.com/watch?v=eCj0cRtJvJg) <<https://www.youtube.com/watch?v=eCj0cRtJvJg>>

During lab on Wednesday, the instructors will assist you in eluting the carbonyls that are trapped on the cartridges using acetonitrile and then injecting the resulting solution into a high performance liquid chromatograph (HPLC) with UV/Vis detector. Discuss the methods and instrumentation with the instructors so that appropriate detail can be provided in your laboratory report, including the mobile and stationary phase, the mode of HPLC operation, and the mode of detection, and the appropriate instrument calibrations. Pay close attention to the instruction you are provided for using measurements of standards to relate the retention times (x-axis of the chromatogram) and intensities (y-axis) to the types and abundances of the set of carbonyl compounds present.

Resources

DNPH Manual ([link](https://drive.google.com/file/d/1kIUxEaHtLolt1s3gba3s77RTZVa4IFlb/view?usp=sharing)) <<https://drive.google.com/file/d/1kIUxEaHtLolt1s3gba3s77RTZVa4IFlb/view?usp=sharing>>

HPLC Basics ([link](https://www.thermofisher.com/us/en/home/industrial/chromatography/chromatography-learning-center/liquid-chromatography-information/hplc-basics.html)) <<https://www.thermofisher.com/us/en/home/industrial/chromatography/chromatography-learning-center/liquid-chromatography-information/hplc-basics.html>>

Estimating contributions of indoor and outdoor sources to indoor carbonyl concentrations ([link](https://www.sciencedirect.com/science/article/pii/S1352231005011672)) <<https://www.sciencedirect.com/science/article/pii/S1352231005011672>>

Lab 3. Carbon Mass Balance

🔗 Learning Objectives

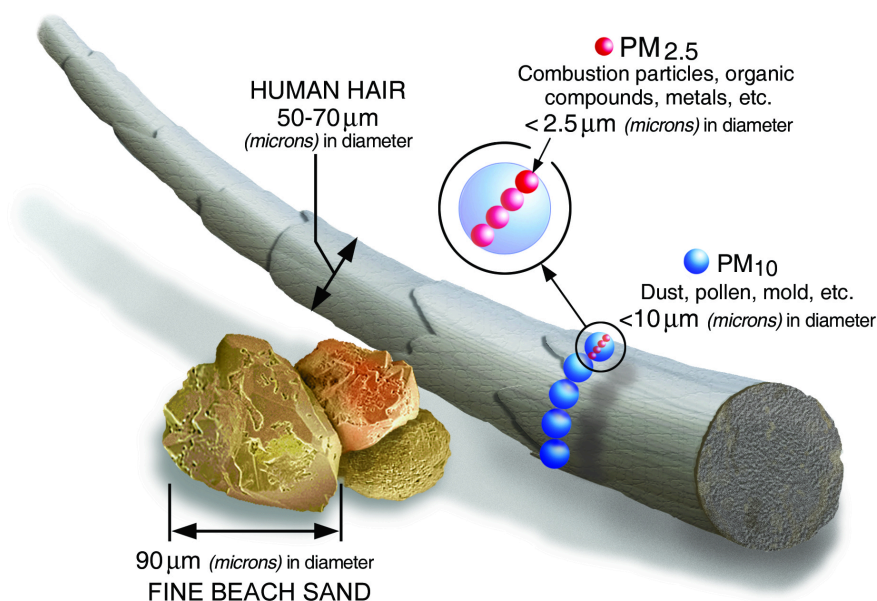
- Explain the principle of operation for aerosol measurements.
- Explain the principle of operation for chip-based gas measurements.
- Perform emission factor measurements from combustion sources using the carbon mass balance method.
- Evaluate the emission factor on a per fuel basis in the context of the peer-reviewed literature.

Required Reading

Introduction to [Combustion Emissions](https://mdpetters.github.io/enve160b/magic/combustion.pdf) <<https://mdpetters.github.io/enve160b/magic/combustion.pdf>>.

Background

PM stands for particulate matter (also called particle pollution): the term for a mixture of solid particles and liquid droplets found in the air. Some particles, such as dust, dirt, soot, or smoke, are large or dark enough to be seen with the naked eye. Others are so small they can only be detected using an electron microscope. Particle pollution includes PM_{10} , which are inhalable particles, with diameters that are generally 10 micrometers and smaller and $PM_{2.5}$, which are fine inhalable particles, with diameters that are generally 2.5 micrometers and smaller. (Source: [EPA](https://www.epa.gov/pm-pollution/particulate-matter-pm-basics) <<https://www.epa.gov/pm-pollution/particulate-matter-pm-basics>>).



Particulate matter is directly emitted in the form of smoke from combustion processes. This includes (wild)fires, residential and commercial cooking, and combustion engines. The size distribution and chemical composition of the formed particulate matter is complex and depends on the fuel and combustion process. A first estimate of the source strength of a process is the emission factor.

An emissions factor is a representative value that attempts to relate the quantity of a pollutant released to the atmosphere with an activity associated with the release of that pollutant. These factors are usually expressed as the weight of pollutant divided by a unit weight, volume, distance, or duration of the activity emitting the pollutant (e.g., kilograms of particulate emitted per megagram of coal burned). Such factors facilitate estimation of emissions from various sources of air pollution. In most cases, these factors are simply averages of all available data of acceptable quality, and are generally assumed to be representative of long-term averages for all facilities in the source category (i.e., a population average). (Source: EPA <<https://www.epa.gov/air-emissions-factors-and-quantification/basic-information-air-emissions-factors-and-quantification>>).

In the atmosphere or indoor environment, air often dilutes by mixing. Emission factors can be estimated using the simplified carbon balance method, which assumes that all carbon in the fuel is converted to CO_2 . In this method the change in observed property (ozone O_3 , particle number N , or particle mass $PM_{2.5}$) is normalized by the change in CO_2 . The increase in CO_2 serves as a measure of fuel combusted and accounts for the dilution with background air. (During combustion the vast majority of carbon is converted to CO_2). An example is shown below.

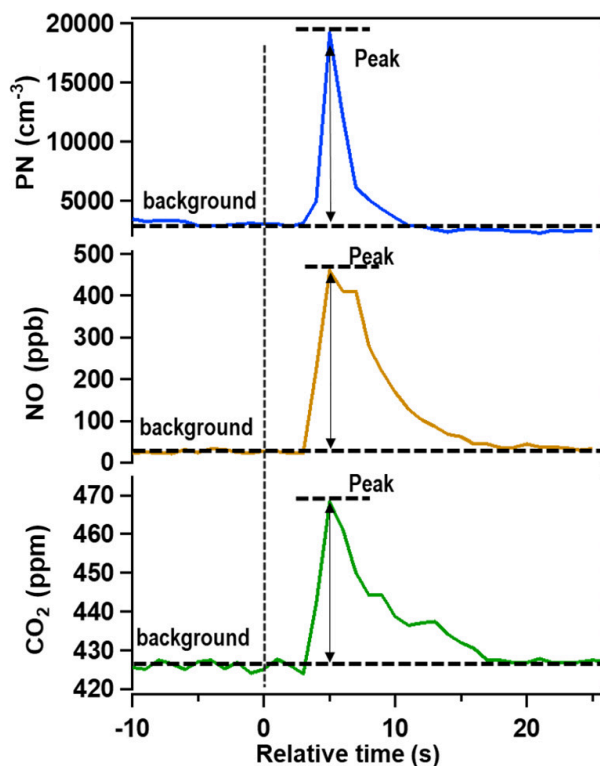


Figure (Source: Shen et al., 2022 <<https://doi.org/10.1016/j.scitotenv.2021.151609>>). Time series of pollutant signals. The vertical dashed line indicates the time at which a vehicle passes the sensor platform.

The figure shows a time series of particle number concentration, NO , and CO_2 . The dashed horizontal line is the baseline concentration. In the plume, the particle number concentration increased from $4,500\text{ cm}^{-3}$ (baseline) to $19,000\text{ cm}^{-3}$ (peak plume conditions). The change is $\Delta N = N_{\text{plume}} - N_{\text{baseline}} = 19,000 -$

$4,500\text{ cm}^{-3} = 14,500\text{ cm}^{-3}$. At the same time, CO_2 increased from 427 ppm to 470 ppm. The change is $\Delta CO_2 = CO_{2,plume} - CO_{2,baseline} = 470 - 427\text{ ppm} = 43\text{ ppm}$. Based on these changes, the emission factor can be calculated using the carbon mass balance method, which is described in detail in the required reading.

Prompt

The objective is to measure the emission factor of a number of gases and particulate matter from combustion sources of your choice. You will take a set of chip-based sensors bundled in the UCR Air Quality Kiosk. Expose the Kiosk to emission. Examples include smoke from candles, incense, vehicle emissions near a tailpipe (e.g. in a parking lot or from a bus), or cooking emissions from a grill. You will carry those sensors to a suitable site. First, you will measure and record the background concentrations far from the plume. You will need to carefully consider how you define background conditions. Next, you will try to get close to the source where you can observe an increase in CO and/or CO_2 above background. Finally, place the remaining sensors to get a measurement of the concentrations in the plume. You should familiarize yourself with the sensors on Monday and evaluate some practice targets. You may take the Kiosk home and target emissions there if you like. Discuss your preliminary results with the instructor and/or TA. Show preliminary data analysis on Wednesday and adjust the measurement protocol based on lessons learned.

Report

Please follow the [report outline](#)

https://docs.google.com/document/d/1PhQsRvCM1_sqDcy18_a6RG8RHiOktbe_sn7TnOJm_G4/edit?usp=sharing.

Don't forget to print and paste the [grading rubric](#)

https://docs.google.com/document/d/1ocOcguu2LU4kSziEKwV7fkrkh7znVgPouguTiv_PYms/edit?usp=sharing at the end of your report. Not providing the grading rubric will lead to point deductions. Use the grading rubric as a checklist for how to prepare a proper report.

Methods

The UCR Air Quality Kiosk contains chip-based sensors for PM , CO_2 , CO , $HCHO$ (formaldehyde), SO_2 , H_2S , NO , and NO_2 . Details about these sensors are provided in the required reading. During your experiment, not all sensors may show measurable signal. Make sure that you are measuring in regimes that do not saturate the sensor. Data on the Kiosk are stored to a USB disk and are reported as a comma separated data file. You can remove the USB stick, copy the data file, and then use computer software to load the time series and plot the data.